Video-Text Multi-Instance Retrieval in Egocentric videos: Developing a Model for Semantic Alignment for the EPIC-KITCHENS-100 Challenge

María Castro Iregui Santiago Martínez Novoa
Nicolás Rozo Fajardo Juan Sebastián Urrea López

1 Context and Problem

Current situation. The growth of audiovisual content has driven the development of methods capable of analyzing, indexing, and automatically retrieving visual information. In this context, multimodal video-text retrieval has established itself as a relevant area at the intersection of computer vision and natural language processing. An especially challenging scenario for these tasks is that of egocentric video, which captures activities from a first-person perspective using body- or head-mounted cameras. This type of video records natural interactions between hands, objects, and the environment, introducing high visual and temporal variability. Consequently, models must be able to interpret actions and contextual relationships in order to establish precise correspondences between visual content and its textual description. One of the most relevant resources for this problem is the *EPIC-KITCHENS-100* dataset, which contains over 100 hours of egocentric video in real domestic environments and approximately 90 000 annotated segments. The *EPIC-KITCHENS-100 Multi-Instance Retrieval Challenge* is built on this dataset; its goal is to align video clips and textual descriptions through relevance matrices with soft labels, allowing different degrees of semantic similarity between video–text pairs to be modeled.

Problem and relevance. Despite recent advances in multimodal models, semantic alignment between video and natural language remains challenging, particularly in egocentric scenarios. The dynamics of these videos—characterized by constant perspective changes, occlusions, and multiple object interactions—make it difficult to extract robust visual representations. Furthermore, in multimodal retrieval tasks a single video may be related to multiple descriptions at different levels of similarity. This requires models capable of estimating continuous semantic relations between video and text, going beyond approaches based solely on binary correspondences. It is therefore necessary to develop methods that integrate visual and textual representations in a shared semantic space, enabling more precise modeling of the relationships between observed actions and natural language descriptions.

Expected impact. This project will contribute to the study of deep learning methods for semantic alignment between video and natural language in egocentric scenarios. The results will allow the exploration of strategies to generate multimodal representations capable of modeling different degrees of similarity between both modalities. Advances in these methods can facilitate the development of search and automatic organization systems for large video repositories, as well as applications in intelligent assistants based on egocentric vision and automated human activity analysis.

2 General Objective

To develop and evaluate a deep learning model for multimodal video-text retrieval using the *EPIC-KITCHENS-100* dataset, capable of identifying shared semantic representations that align egocentric video segments and textual descriptions, thereby generating relevance rankings for both video-to-text and text-to-video retrieval tasks.

3 Specific Objectives

1. To analyze the structure of the *EPIC-KITCHENS-100* dataset and the relevance matrices used in the multimodal retrieval challenge, identifying the organization of video segments, textual descriptions, and their semantic relationships.

2. To design a deep learning architecture that integrates temporal visual representations and semantic language representations to project both modalities into a shared feature space.

3. To implement and train the proposed model to estimate semantic similarity between video–text pairs and generate retrieval rankings for both search directions.

4. To evaluate the model's performance using ranking metrics such as *Normalized Discounted Cumulative Gain* (nDCG) and *Mean Average Precision* (mAP), comparing the results against baseline approaches reported in the literature.

4 State of the Art

Multimodal video-text retrieval has seen accelerated advances in recent years, driven by the convergence of three research lines: contrastive learning, vision-language foundation models, and first-person video understanding. This section reviews the most relevant works organized around three thematic axes: (i) adaptation of vision-language models for video retrieval, (ii) multimodal pretraining in the egocentric domain, and (iii) multi-instance retrieval with graded relevance.

Adaptation of vision-language models for video retrieval. Contrastive image-text pretraining models learn shared representation spaces from large collections of image-text pairs and have laid the groundwork for multimodal retrieval with video. Luo et al. proposed CLIP4Clip, an empirical study that adapts this class of models to video clip retrieval through three temporal aggregation strategies: frame representation averaging, sequential transformers, and tight coupling between

encoders [1]. Their experiments on MSR-VTT, MSVD, and ActivityNet Captions showed that the quality of pretrained representations outweighed the choice of temporal aggregation strategy, reinforcing the central role of the shared representation space. Ma et al. extended this paradigm with X-CLIP, which introduces multi-granularity contrastive learning capable of simultaneously aligning video-sentence pairs at the global level and frame-word pairs at the local level, via a cross-frame communication transformer [2]. This fine-grained correspondence capture represents a notable improvement over single-level alignment methods. In a complementary direction, Rasheed et al. demonstrated with ViFi-CLIP that end-to-end fine-tuning of vision-language models for video, with minimal architectural modifications, achieves competitive results in zero-shot generalization [3], showing that shared embedding spaces have a high transfer capacity to new domains.

Multimodal pretraining in the egocentric domain. First-person video presents particular challenges: dynamic camera motion, high intra-class variability in everyday actions, and a data distribution very different from third-person video. The corpus that enabled large-scale pretraining in this domain is Ego4D, collected by Grauman et al. in a collaboration of thirteen universities and Meta AI [4]. With over 3,600 hours of first-person video recorded by 931 participants in nine countries, Ego4D established the data infrastructure on which the most recent egocentric models are built. Damen et al. introduced EPIC-KITCHENS-100, the reference dataset for egocentric cooking activities, which also establishes the multi-instance retrieval (MIR) task based on relevance matrices with soft labels [5]. This formulation recognizes that multiple video segments may be relevant to the same textual query at different degrees of semantic similarity, requiring order-sensitive evaluation metrics (such as nDCG and mAP) rather than the traditional R@K.

To capitalize on the availability of the Ego4D corpus, Lin et al. presented EgoVLP, the first video-language pretraining model designed specifically for the egocentric domain [6]. EgoVLP introduces EgoNCE, a variant of InfoNCE that weights additional positive pairs through verb and noun matching, and achieves substantial improvements over general-domain methods on EK-100. However, the handling of multiple instances in EgoNCE relies on lexical overlap rather than directly leveraging the official graded relevance matrices of the challenge. Zhao et al. advanced in this direction with LAVILA, which employs a language model fine-tuned on Ego4D to generate diverse narrations of training clips, thereby enriching the available semantic supervision [7]. LAVILA achieved state-of-the-art results on EK-100 MIR at the time of publication and became the reference video encoder for subsequent works. Pramanick et al. proposed EgoVLPv2, which integrates cross-modal fusion directly into the network backbone via cross-attention layers interleaved with self-attention layers [8], obtaining richer joint representations by capturing temporal and semantic dependencies in an integrated manner. Building on LAVILA's encoder design, Zhao and Krähenbühl introduced AVION, a framework for efficient large-scale egocentric video-language pretraining that trains a ViT-L/14 backbone on a single machine in a single day [9]. AVION won the CVPR 2023 EK-100 challenge and its ViT-L/14 encoder has since become a widely adopted backbone for downstream retrieval work.

Multi-instance retrieval, graded relevance, and foundation models. The official EK-100 MIR challenge baseline combines two models provided by the organizers. The first is JPoSE, proposed by Wray et al., which learns separate embedding spaces for verbs, nouns, and full actions, combining them for fine-grained retrieval [10]. The second is MI-MM, an adaptation of MIL-NCE [11] with a multi-instance MaxMargin loss and S3D features, which serves as an explicit comparison point for the challenge. Both models operate with binary relevance objectives and do not exploit the soft-label matrices.

Scaling towards larger video foundation models has pushed the limits of multimodal understanding. Wang et al. presented InternVideo2, which combines masked video pretraining, contrastive visual-textual alignment, and instruction-based fine-tuning, achieving state-of-the-art results on more than sixty evaluation benchmarks [12]. In the specific context of the EK-100 MIR challenge, Wang, Wang, and Chau proposed a symmetric multiple similarity loss that incorporates the soft-label relevance matrices directly into the cost computation, applied symmetrically over visual and textual representations [13]. This approach, built on LAVILA as the base encoder, represents one of the most explicit attempts to directly optimize the nDCG and mAP metrics of the competition.

In parallel, the visual retrieval community has developed loss functions that directly optimize ranking metrics. Engilberge et al. proposed SoDeep, an auxiliary network that learns to approximate the sorting operation to make mAP and Recall computation differentiable, successfully applying it to cross-modal text-image retrieval [14]. Brown et al. introduced Smooth-AP, a sigmoidal approximation of Average Precision that enables its direct large-scale optimization in image retrieval [15]. Despite these advances, this family of methods has not been systematically explored in the context of video-text retrieval with graded relevance, which represents a direct opportunity for the present work. More recently, He et al. proposed CR-CLIP (ContextRefine-CLIP), which builds on the AVION dual-encoder architecture and introduces context refinement mechanisms over pretrained CLIP representations for multi-instance retrieval [16]. CR-CLIP achieved first place on the EK-100 MIR 2025 public leaderboard with a mean nDCG of 82.1% and mAP of 66.8%, establishing the current state of the art on the benchmark.

Identified gaps and project focus. Despite the advances described, important limitations remain. Most video-text retrieval methods were designed for third-person video with binary relevance annotations, without considering the multi-instance structure of the egocentric scenario. Models specialized for first-person video, such as EgoVLP and LAVILA, employ loss functions that only partially leverage the graded relevance matrices: EgoNCE uses lexical overlap as a surrogate for true semantic similarity, while LAVILA applies standard contrastive learning without distinguishing degrees of relevance. Approaches that do incorporate soft labels, such as that of Wang et al. [13], do so as a fine-tuning stage over representations pretrained with binary objectives. This project seeks to close this gap by developing a multimodal retrieval model

that learns shared semantic representations for egocentric video segments and textual descriptions, explicitly integrating the soft-label relevance matrices from the training process itself, with the goal of directly maximizing the nDCG and mAP metrics of the EPIC-KITCHENS-100 MIR challenge.

5 Methodology and Proposed Focus

5.1 Technical focus

This project will begin with a thorough analysis of the dataset, presented in the following section. During this stage, we will identify its structure, composition and possible pre-processing filters that can be applied to enhance the training data.

Then, we will begin with the design of an ML/DL architecture based on the foundational models established in EgoVLP [6], LAVILA [7] and JPoSE [10]. Starting off from foundational models proves useful in fine-tuning and will enable an exploration of the impact of processing in multimodal retrieval. This architecture uses a dual-encoder projected on a shared representation space and SMS-Loss over the soft-label relevance matrices as the loss function.

To implement and experiment with this architecture, we will use a modern and robust MLOps pipeline that pulls weights from AVION ViT-L/14 [9], SMS-Loss [13] and CR-CLIP [16], the two previous winner build on the aforementioned architecure. Experiment management will be done with HYDRA and composable YAML files. Due to the GPU-intensiveness of these models, this implementation will be deployed on the cloud and potentially integrate into Universidad de los Andes' cluster.

Overall, this pipeline will enable quick experimentation and provide an organized structure for the analysis of the metrics of interest (nDCG and mAP) in order to select and iterate on the best results.

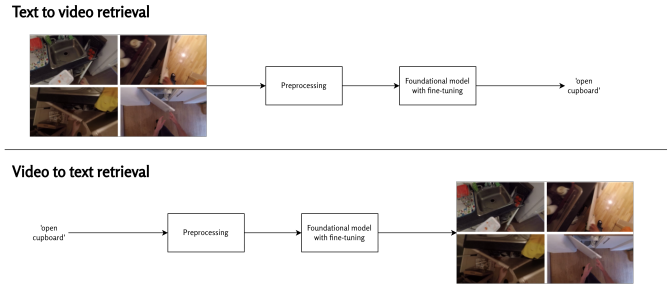

Figure 1: Diagram of solution with expected inputs an outputs. Images adapted from [17]

5.2 Libraries and Tools

The following libraries will be used:

```
PyTorch, torchvision, torchaudio (CUDA 12.1)
transformers, timm, einops
hydra-core, omegaconf, pydantic
wandb, tensorboard
matplotlib, seaborn, plotly
```

6 Data Exploration and Description

6.1 Data characteristics

Description of the available data sources along with their profiling and characteristics. If additional images need to be collected, indicate the sources used.

6.2 Exploratory analysis

A notebook with the full exploration must be attached.

- Average pixel values
- Distribution of each image: size
- Resolution
- Plots
- Image examples
- Video metadata (duration, resolution, fps)
- Average video length

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